Faculty of Computing

BSc (Hons) in Software Engineering
Year 2 Semester 2 (2026)

SE2020 – Web and Mobile Technologies Group Assignment

Group Assignment: Full Stack Mobile Application
Weight: 20% (Marked out of 100 and scaled)

Group Size: 6 Students
Duration: 8 Weeks

1. Assignment Overview

Students must design and develop a Full Stack Mobile Application using:

- **Frontend:** React Native
- **Backend:** Node.js + Express.js
- **Database:** MongoDB

The system must be deployed to any hosting platform (AWS, Render, Railway, DigitalOcean, etc.).

Students may choose any system title/domain, but **approval is required**.

2. Core System Requirements (Mandatory)

Every group project MUST include:

1. **User Authentication**
   - User Registration
   - Login
   - Password hashing
   - JWT-based authentication
   - Protected routes

2. **Hosting Requirement**
   - Backend must be hosted online
   - Mobile app must connect to hosted API
   - Live demo must show deployed version
   - (Localhost demo not allowed for final evaluation)

3. Workload Distribution (Equal for 6 Members)

Each member MUST handle a clearly defined module.

Group – Authentication
Responsible for:

- User registration API
- Login API
- Password hashing
- JWT-based authentication

Each Member – Core Entity CRUD
Responsible for:

- Full CRUD backend for main entity
- File upload
- Mobile UI front end
- API routes
- Controllers
- Integration with database
- Testing

Example: Product Management (add/remove/update/delete products) or Task Management / Appointment Management System / User Management

4. Technical Requirements

Backend Requirements

- RESTful APIs
- Proper folder structure
- Middleware usage
- Error handling
- Status codes

5. Mobile Requirements

- Proper navigation
- Functional components & hooks
- Clean UI

- Form validation
- API integration
- No hardcoded data

Must include:

- Problem statement
- System architecture diagram
- Database schema diagram
- API endpoint table
- Team responsibility breakdown

6. Important Rules

- All 6 members must attend viva.
- If a student cannot explain their module → individual marks reduced.
- Plagiarism equals Zero marks.
- AI tools are allowed for learning support, not full system generation.
- No Firebase-only projects allowed (must use Node + MongoDB backend).

7. Example Project Ideas (Optional)

- Event Booking App
- Clinic Appointment System
- Food Ordering App
- Hostel Management App
- E-commerce Mobile App
- Fitness Tracking System

You can use the same project that you are using for ITP project / ML&AI projects.

Example Project
Hostel Management Mobile Application
(React Native + Node.js + MongoDB + Hosted Backend)

Core System Overview

The system will allow:

- Students to register & login
- Admin to manage rooms
- Students to request rooms
- Admin to approve/reject bookings
- Image upload for room photos

Main Entities

Entity 1: Room
This is the primary entity
Fields:

- `roomNumber`
- `roomType` (Single / Double / Triple)
- `pricePerMonth`
- `capacity`
- `currentOccupancy`
- `description`
- `image`
- `availabilityStatus`

CRUD Required:

- Create Room (Admin)
- View Rooms (All users)
- Update Room (Admin)
- Delete Room (Admin)

Entity 2: Booking
This links users to rooms.
Fields:

- `userId` (reference to User)
- `roomId` (reference to Room)
- `bookingDate`
- `startDate`
- `endDate`
- `status` (Pending / Approved / Rejected)

CRUD Required:

- Create booking request
- View booking history
- Update status (Admin approval)
- Cancel booking

Student	Entity	Focus Area
Student 1	Room	Room management
Student 2	Booking	Room Booking
Student 3	Staff	Staff Management
Student 4	Payment	Financial Management
Student 5	Complaint	Maintenance System
Student 6	Visitor	Security & Visitor Tracking

Member 1 – Entity 1 Lead (Room Management API)

Responsible for:

- Full CRUD for Room
  - Create room (Admin)
  - Get all rooms
  - Get single room
  - Update room
  - Delete room

Business Logic:

- Update availability status
- Adjust current occupancy

Viva Focus:

- Explain controller logic
- Explain route design
- Explain data validation

Technical Responsibilities:

- Navigation setup
- API integration
- Form validation
- State management
- Error message display

Member 6 – Image Upload & Deployment Engineer

Responsibilities:

Image Upload:

- Upload room image (Multer)
- Validate file type & size

- Store in server/cloud
- Send image URL to frontend

Deployment:

- Deploy backend (AWS / Render / Railway)
- Configure environment variables
- Ensure MongoDB Atlas connection
- Test live API endpoints

Viva Focus:

- Explain deployment steps
- Explain environment config
- Explain image handling pipeline

Individual Mark Sheet (Common for All 6 Members)

A. Code / Technical Implementation – 40 Marks

(Assessed from Git + System Testing + Module Ownership)

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B. Individual Viva – 60 Marks
(Each student questioned individually)

Criteria	Description	Marks
Explaining Own Module in Detail	Can explain logic, functions, flow clearly	20
Explain Integration with Other Modules	Understands how system connects	10
Backend & Database Concepts	Understands schema, routes, controllers	10
Mobile & API Interaction Logic	Understands request flow & UI behavior	10
Problem Solving & Debugging	Can answer "what if" and fix scenario questions	10
Total		**60**